POLITECNICO DI TORINO

Lab 1 Exercise 1 2: FFmpeg

Audio Analysis

Complete Report

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1 Introduction

The purpose of this exercise is to work with audio compression formats through FFmpeg and to evaluate the effect of compression on an audio signal. The original input file used in the exercise is `dogs.wav`. The assignment requires first encoding this file into six different compressed versions, then decoding all compressed files back to uncompressed WAV, and finally comparing the reconstructed signals against the original one using both objective and perceptual analysis.

The six required compressed files are:

1. MP3 – CBR at 320 kbps
2. MP3 – CBR at 96 kbps
3. MP3 – VBR maximum quality
4. Opus at 256 kbps
5. Opus at 48 kbps
6. FLAC with maximum compression level

After generating these files, the assignment requires the following analysis steps:

1. Convert the six encoded files back to uncompressed WAV.
2. Prove that the FLAC-compressed version is identical to the original sample-wise.
3. Excluding the sixth file, compute:
   - Mean Absolute Error (MAE),
   - Mean Squared Error (MSE),
   - Signal-to-Noise Ratio (SNR).
4. Generate the error files for file 1 and file 5.
5. Listen to the generated error files and report comments on them.

Therefore, the exercise combines the practical use of FFmpeg with a quantitative signal analysis of lossy and lossless compression.

2 Software environment and tools

The exercise was carried out in the CrownLabs Jupyter environment using the following tools:

- **FFmpeg** for audio encoding and decoding,
- **Python** for numerical analysis,
- **NumPy** for vectorized operations on audio samples,
- **SciPy** for reading and writing WAV files,
- **Pandas** for organizing the numerical results,
- **Matplotlib** in the notebook for visualizing the SNR values.

The original file used in the exercise was:

```
dogs.wav
```

3 Part 2.1 – Using FFmpeg

3.1 Required encodings

The original WAV file was encoded into six different compressed formats exactly as required by the assignment. The command strings used were the following:

FFmpeg Encoding Commands

```
ffmpeg -y -i dogs.wav -c:a libmp3lame -b:a 320k dogs_1_mp3_cbr_320.mp3
ffmpeg -y -i dogs.wav -c:a libmp3lame -b:a 96k  dogs_2_mp3_cbr_096.mp3
ffmpeg -y -i dogs.wav -c:a libmp3lame -q:a 0    dogs_3_mp3_vbr_max.mp3
ffmpeg -y -i dogs.wav -c:a libopus    -b:a 256k dogs_4_opus_256.opus
ffmpeg -y -i dogs.wav -c:a libopus    -b:a 48k  dogs_5_opus_048.opus
ffmpeg -y -i dogs.wav -c:a flac -compression_level 12 dogs_6_flac_max.flac
```

3.2 Explanation of the commands

Each command follows the general FFmpeg structure:

```
ffmpeg -y -i input_file [codec settings] output_file
```

where:

- `-y` forces overwriting of the output file if it already exists,

- `-i dogs.wav` specifies the input audio file,
- `-c:a` selects the audio codec,
- `-b:a` sets the target audio bitrate,
- `-q:a 0` sets the highest VBR quality for MP3,
- `-compression_level 12` sets the maximum FLAC compression level.

The required codecs are exactly those specified by the assignment:

- `libmp3lame` for MP3,
- `libopus` for Opus,
- `flac` for FLAC.

3.3 Meaning of each encoding

File 1: MP3 CBR 320 kbps This is a high-bitrate constant-bitrate MP3 encoding. Since 320 kbps is the maximum bitrate requested in the assignment, this file is expected to preserve the original signal relatively well while still being lossy.

File 2: MP3 CBR 96 kbps This is a lower-bitrate MP3 encoding. Because the bitrate is much smaller, a stronger degradation is expected.

File 3: MP3 VBR maximum quality This is a variable-bitrate MP3 encoding in maximum-quality mode. The encoder allocates bits adaptively depending on the complexity of the signal, producing an intermediate trade-off between compression and quality.

File 4: Opus 256 kbps This is a high-bitrate Opus encoding. Since Opus is a modern perceptual codec, this version is expected to preserve the audio reasonably well.

File 5: Opus 48 kbps This is a low-bitrate Opus encoding. Because the bitrate is low, more information must be removed, so stronger distortion is expected.

File 6: FLAC maximum compression This file uses lossless compression. Even though the file is compressed, the decoded signal should remain identical to the original one sample-by-sample.

4 Part 2.2 – Analyzing outputs

4.1 Conversion back to uncompressed WAV

The first required step in the analysis section is to convert the six encoded files back to uncompressed WAV. One FFmpeg command string used to convert back one of the six files to WAV is:

Decoding Command

```
ffmpeg -y -i dogs_1_mp3_cbr_320.mp3 dogs_1_mp3_cbr_320_dec.wav
```

For completeness, the full set of decoding commands used is:

Full Decoding Set

```
ffmpeg -y -i dogs_1_mp3_cbr_320.mp3  dogs_1_mp3_cbr_320_dec.wav
ffmpeg -y -i dogs_2_mp3_cbr_096.mp3  dogs_2_mp3_cbr_096_dec.wav
ffmpeg -y -i dogs_3_mp3_vbr_max.mp3  dogs_3_mp3_vbr_max_dec.wav
ffmpeg -y -i dogs_4_opus_256.opus    dogs_4_opus_256_dec.wav
ffmpeg -y -i dogs_5_opus_048.opus    dogs_5_opus_048_dec.wav
ffmpeg -y -i dogs_6_flac_max.flac    dogs_6_flac_max_dec.wav
```

4.2 Loading the original and decoded WAV files

The original file and the six decoded WAV files were loaded into the notebook for analysis. The output obtained from the notebook execution was:

```
Original:  48000 (1364219,) int16
1 MP3 CBR 320:  48000 (1364219,) int16
2 MP3 CBR 96:  48000 (1364219,) int16
3 MP3 VBR max:  48000 (1364219,) int16
4 Opus 256:  48000 (1364219,) int16
5 Opus 48:  48000 (1364219,) int16
6 FLAC max:  48000 (1364219,) int16
```

This means that:

- the sample rate of the original file is 48000 Hz,
- the sample rate of every decoded file is also 48000 Hz,
- each signal contains 1364219 samples,
- all signals are stored as `int16`.

4.3 Verification that all signals are directly comparable

Before computing any metric, the notebook checked that all files had matching sample rates and shapes. The output was:

```
All sample rates match.
All shapes match.
```

This verification is essential because all subsequent comparisons assume that corresponding samples in the original and reconstructed signals refer to the same positions in time.

5 Proof that the FLAC file is sample-wise identical

The assignment explicitly requires proving that the FLAC-compressed version is identical to the original sample-wise. The notebook performed the following checks:

- equality of the sample rate,
- equality of the signal shape,
- direct sample-by-sample comparison using array equality,
- computation of the maximum absolute difference.

The obtained output was:

```
Same sample rate:  True
Same shape:  True
Sample-wise identical:  True
Maximum absolute difference:  0
```

This proves that if x is the original signal and y_6 is the decoded FLAC signal, then $y_6 = x$ for every sample. Equivalently, the sample-wise difference is exactly zero: $y_6 - x = 0$ for all sample indices. Since the maximum absolute difference is also zero, there is no numerical discrepancy at all between the original and the decoded FLAC version. Therefore, FLAC behaves as a **lossless** codec in this experiment.

6 Definition of the error signal

The assignment provides a hint stating that computations are easier if the *error signal* is computed first, and that this same error signal is needed for the next point. Another hint states that there is no need for `for`-loops. Accordingly, the analysis was performed in vectorized form in Python.

For each decoded lossy file y, the error signal was defined as:

$$e = y - x$$

where x is the original signal and y is the decoded compressed signal. More specifically, for the five lossy files:

- $e_1 = y_1 - x$
- $e_2 = y_2 - x$
- $e_3 = y_3 - x$
- $e_4 = y_4 - x$
- $e_5 = y_5 - x$

The number of samples used in the computation was $N = 1364219$.

7 Mathematical formulas used in the analysis

Mean Absolute Error (MAE)

The Mean Absolute Error is defined as:

$$\text{MAE} = \frac{1}{N} \sum_{i=0}^{N} |y_i - x_i|$$

This metric measures the average magnitude of the absolute difference between the reconstructed and original signals.

Mean Squared Error (MSE)

The Mean Squared Error is defined as:

$$\text{MSE} = \frac{1}{N} \sum_{i=0}^{N} (y_i - x_i)^2$$

This metric squares the error before averaging it, so larger deviations contribute more strongly.

Signal-to-Noise Ratio (SNR)

The Signal-to-Noise Ratio is defined as:

$$\mathrm{SNR} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right)$$

with $P_{\text{noise}} = \mathrm{MSE}$ and $P_{\text{signal}} = \frac{1}{N} \sum_{i=0}^{N} x_i^2$. The SNR measures how large the signal power is compared to the noise power introduced by the codec.

8 Numerical results

The computed values for files 1 to 5 were the following:

Table 1: Computed MAE, MSE, and SNR values for files 1 to 5

File	MAE	MSE	SNR [dB]
`1_MP3_320`	6.102702	76.734200	54.099386
`2_MP3_096`	221.709621	85213.442779	23.644194
`3_MP3_VBR`	69.738377	9258.827239	33.283715
`4_Opus_256`	67.165495	8370.095277	33.721971
`5_Opus_048`	515.549662	465676.278460	16.268434

9 Detailed discussion of the numerical results

9.1 File 1 – MP3 CBR 320 kbps

For file 1, the measured values are:

$$\mathrm{MAE}_1 = 6.102702, \quad \mathrm{MSE}_1 = 76.734200, \quad \mathrm{SNR}_1 = 54.099386 \text{ dB}$$

These are the best values among the lossy encodings: the lowest MAE, the lowest MSE, and the highest SNR. This means that the decoded MP3 at 320 kbps is extremely close to the original signal. The distortion introduced by the codec is very small.

9.2 File 2 – MP3 CBR 96 kbps

For file 2, the values are:

$$\mathrm{MAE}_2 = 221.709621, \quad \mathrm{MSE}_2 = 85213.442779, \quad \mathrm{SNR}_2 = 23.644194 \text{ dB}$$

Compared to file 1, the degradation is much stronger. The lower bitrate forces the encoder to remove more information, and this is reflected in the substantially larger MAE and

MSE and the much lower SNR.

9.3 File 3 – MP3 VBR maximum quality

For file 3, the values are:

$$\mathrm{MAE}_3 = 69.738377, \quad \mathrm{MSE}_3 = 9258.827239, \quad \mathrm{SNR}_3 = 33.283715 \text{ dB}$$

This result is clearly better than MP3 at 96 kbps and clearly worse than MP3 at 320 kbps. The variable-bitrate encoder allocates bits adaptively and produces an intermediate quality level in this experiment.

9.4 File 4 – Opus 256 kbps

For file 4, the values are:

$$\mathrm{MAE}_4 = 67.165495, \quad \mathrm{MSE}_4 = 8370.095277, \quad \mathrm{SNR}_4 = 33.721971 \text{ dB}$$

This file performs slightly better than file 3 because its MAE is slightly smaller, its MSE is slightly smaller, and its SNR is slightly higher. Therefore, in this experiment, `Opus 256 kbps` gives a marginally better reconstruction than `MP3 VBR maximum quality`.

9.5 File 5 – Opus 48 kbps

For file 5, the values are:

$$\mathrm{MAE}_5 = 515.549662, \quad \mathrm{MSE}_5 = 465676.278460, \quad \mathrm{SNR}_5 = 16.268434 \text{ dB}$$

This is the worst result among the lossy encodings: the largest MAE, the largest MSE, and the lowest SNR. This means that the reconstructed waveform produced by `Opus 48 kbps` differs substantially from the original signal.

10 Ranking of the lossy encodings

A quality ranking can be obtained directly from the SNR values:

$$\text{File 1} > \text{File 4} > \text{File 3} > \text{File 2} > \text{File 5}$$

or explicitly:

1. MP3 CBR 320 kbps
2. Opus 256 kbps

3. MP3 VBR maximum quality
4. MP3 CBR 96 kbps
5. Opus 48 kbps

11 Graphical representation of the SNR results

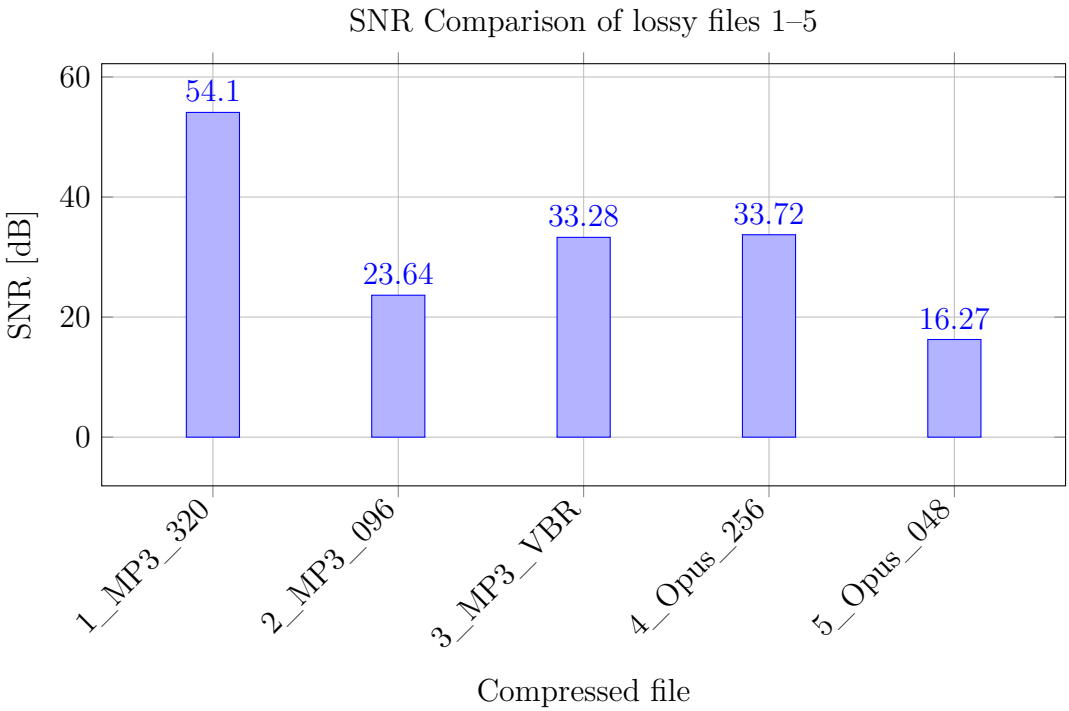

Figure 1: Bar chart of the SNR values for the five lossy compressed files

11.1 Interpretation of the graph

The graph has the five lossy encoded files on the horizontal axis and their SNR values in decibels on the vertical axis. A higher bar corresponds to a lower distortion level and therefore better reconstruction quality. The graph shows the following:

- The bar for `1_MP3_320` is the tallest, reaching approximately 54.10 dB.
- The bar for `2_MP3_096` is much lower, at approximately 23.64 dB.
- The bar for `3_MP3_VBR` is around 33.28 dB.
- The bar for `4_Opus_256` is around 33.72 dB and is slightly higher than file 3.
- The bar for `5_Opus_048` is the shortest, at approximately 16.27 dB.

12 Generation of the error files

The assignment explicitly requires generating the error files for file 1 and file 5. The error signals were defined as $e_1 = y_1 - x$ and $e_5 = y_5 - x$. These signals were written as WAV files:

- `error_file_1_mp3_320.wav`
- `error_file_5_opus_048.wav`

The notebook output confirmed their creation:

```
Created:
error_file_1_mp3_320.wav
error_file_5_opus_048.wav
```

13 Listening analysis of the error files

The last required step of the assignment is to listen to the error files and comment on them.

13.1 Comments on file 1 (MP3 320 kbps)

The comment reported in the notebook was:

> **Comments on file 1 (MP3 320 kbps):** the error signal is expected to be weak and only slightly audible, because the compressed version is usually very close to the original.

This interpretation is consistent with the numerical results, because file 1 has the best values among the lossy encodings. Therefore, the distortion introduced by this codec setting is minimal.

13.2 Comments on file 5 (Opus 48 kbps)

The comment reported in the notebook was:

> **Comments on file 5 (Opus 48 kbps):** the error signal is expected to be more audible, indicating a larger distortion introduced by compression at the lower bitrate.

This is again consistent with the numerical analysis, because file 5 has the worst values among the lossy encodings. Therefore, the codec at this bitrate alters the signal much more strongly.

14 Relationship between objective metrics and perceptual analysis

One important aspect of this exercise is that it combines objective and perceptual evaluation. The objective analysis showed that:

- file 1 has the smallest error and the highest SNR among the lossy encodings,
- file 5 has the largest error and the lowest SNR.

The listening analysis follows the same tendency: the error signal for file 1 is weak, while the error signal for file 5 is more audible. This agreement confirms that MAE, MSE, and SNR are meaningful measures of the amount of distortion.

15 Final discussion

This experiment clearly distinguishes between lossless compression and lossy compression.

15.1 Lossless compression

FLAC belongs to the lossless category. The decoded FLAC file was proven to be sample-wise identical to the original: $y_6 = x$ with maximum absolute difference equal to zero.

15.2 High-quality lossy compression

The following files produced relatively good reconstructions:

- MP3 CBR 320 kbps,
- MP3 VBR maximum quality,
- Opus 256 kbps.

Among these, MP3 CBR 320 kbps gave the best result in this experiment.

15.3 Stronger lossy compression

The lower-bitrate files (`MP3 CBR 96 kbps` and `Opus 48 kbps`) showed much larger errors and lower SNR values. In particular, Opus 48 kbps produced the strongest degradation among all lossy cases.

16 Conclusion

The exercise was completed according to all the points required by the assignment:

1. `dogs.wav` was encoded into six compressed files using FFmpeg.
2. The six compressed files were decoded back to WAV.
3. The decoded FLAC file was proven to be sample-wise identical to the original.
4. The error signals for files 1 to 5 were computed.
5. MAE, MSE, and SNR were computed using the formulas given in the assignment.
6. The error files for file 1 and file 5 were generated.
7. The error files were listened to and commented on.

The final results show that **FLAC** is perfectly lossless, **MP3 320 kbps** is the best lossy encoding, and **Opus 48 kbps** is the most distorted. The numerical metrics, the graph, and the listening comments all support the same conclusion: increasing compression generally increases distortion.